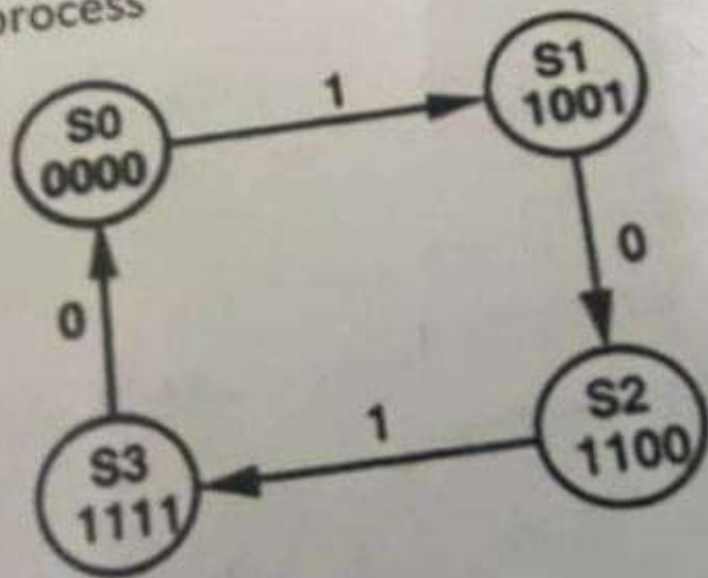
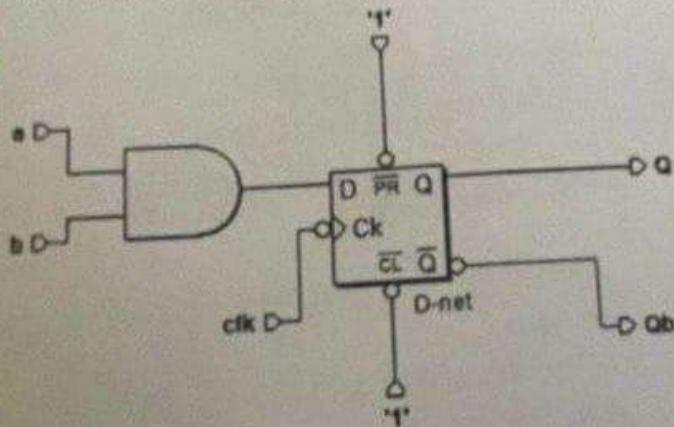


Q3) write the vhdl code for the following FSM machine using the process



Q1. Write the corresponding VHDL code, for the Following circuit?



Q4. Correct the syntax error for the Following vhdl code?

Q5- Draw the FSM for the current VHDL code?

Number of error	

```
library IEEE;
use IEEE.std_logic_1164.all;

entity fsm is
  port ( clk, reset, x1 : IN std_logic;
        outp : OUT std_logic);
end entity;
architecture beh1 of fsm is
  type state_type is (s1,s2,s3,s4);
  signal state: state_type ;
begin
  process (clk,reset)
  begin
    if (reset = '1') then
      state <= s1; outp <= '1';
    elsif (clk = '1' and clk'event) then
      case state is
        when s1 => if x1 = '1' then
state <= s2;
                    else      state <= s3;
                    end if;
                    outp <= '1';
        when s2 => state <= s4; outp
<= '1';
        when s3 => state <= s4; outp <=
'0';
        when s4 => state <= s1; outp <=
'0';
      end case;
    end if;
  end process;
end beh1;
```

2- Draw the logic circuit for the current VHDL code?

```
LIBRARY ieee;
USE ieee.std_logic_1164.all;
USE ieee.numeric_std.all;
ENTITY aaa IS
  PORT(
    S1: IN std_logic;
    S2: IN std_logic;
    S3: IN std_logic;
    d0,d1,d2,d3,d4,d5,d6,d7: OUT
    std_logic );
END aaa;
ARCHITECTURE structural OF aaa IS
Components:
  COMPONENT NOT_gate IS
    PORT( I: IN std_logic;
           O: OUT std_logic );
  END COMPONENT;
  --
  COMPONENT AND3_gate IS
    PORT( I0,I1,I2: IN std_logic;
           O: OUT std_logic );
  END COMPONENT;
  --
  SIGNAL S001: std_logic;
  SIGNAL S002: std_logic;
  SIGNAL S003: std_logic;
  BEGIN -- structural
```

Component Mapping:

```
C073: NOT_gate PORT MAP ( S1, S001 );
C072: NOT_gate PORT MAP ( S2, S002 );
C074: NOT_gate PORT MAP ( S3, S003 );
```

```
c141: AND3_gate PORT MAP ( S001, S002,
S003,d0 );
C142: AND3_gate PORT MAP ( S001, S002,
S3,d1 );
C143: AND3_gate PORT MAP ( S001, S2,
S003,d2 );
C144: AND3_gate PORT MAP ( S001, S2,
S3,d3 );
C145: AND3_gate PORT MAP ( S1, S002,
S003,d4 );
C146: AND3_gate PORT MAP ( S1, S002,
S3,d5 );
```

```
C147: AND3_gate PORT MAP ( S1, S2,
S003,d6 );
C148: AND3_gate PORT MAP ( S1, S2,
S3,d7 );
END structural;
library ieee;
use ieee.std_logic_1164.all;

ENTITY AND3_gate IS
  PORT( I0,I1,I2: IN std_logic;
         O: OUT std_logic );
END AND3_gate;

ARCHITECTURE behavioral OF AND3_gate IS
BEGIN
  O <= (I0 and I1 and I2 );
END behavioral;

library ieee;
use ieee.std_logic_1164.all;

ENTITY NOT_gate IS
  PORT( I: IN std_logic;
         O: OUT std_logic );
END NOT_gate;

ARCHITECTURE behavioral OF NOT_gate IS
BEGIN
  O <= (not I);
END behavioral;
```